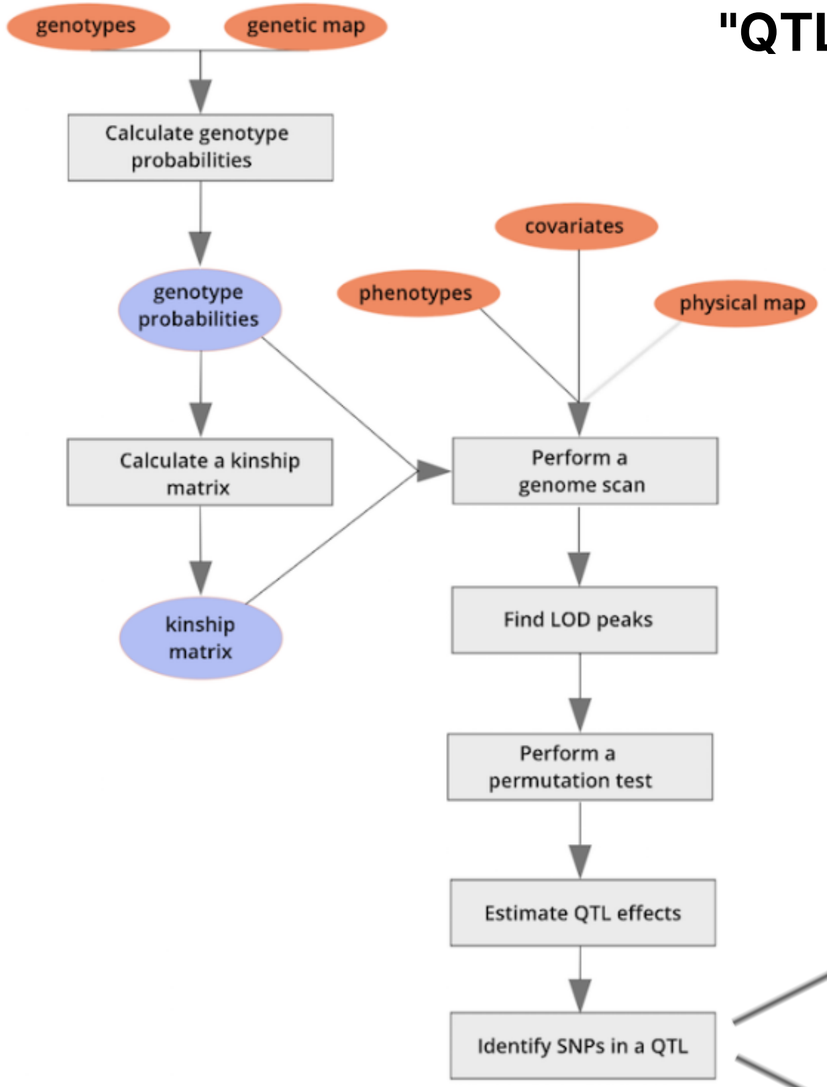
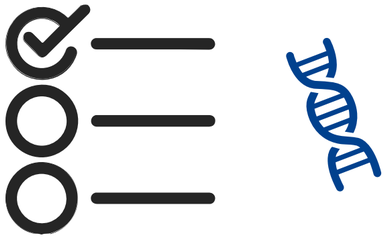
Pheno2GeneRank

"QTL"



Candidate Gene List



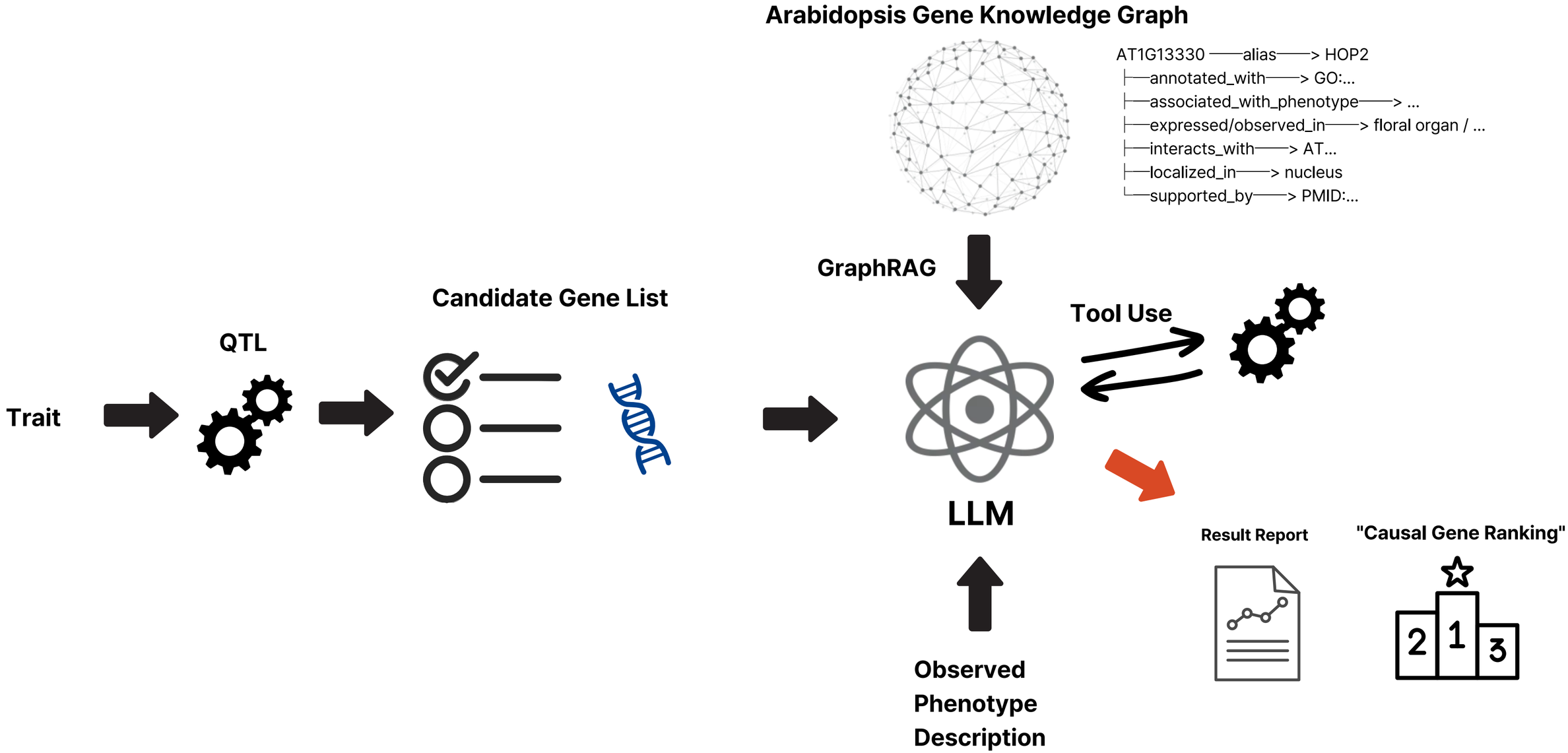
Trait / Phenotype







Gene Exploration & Selection

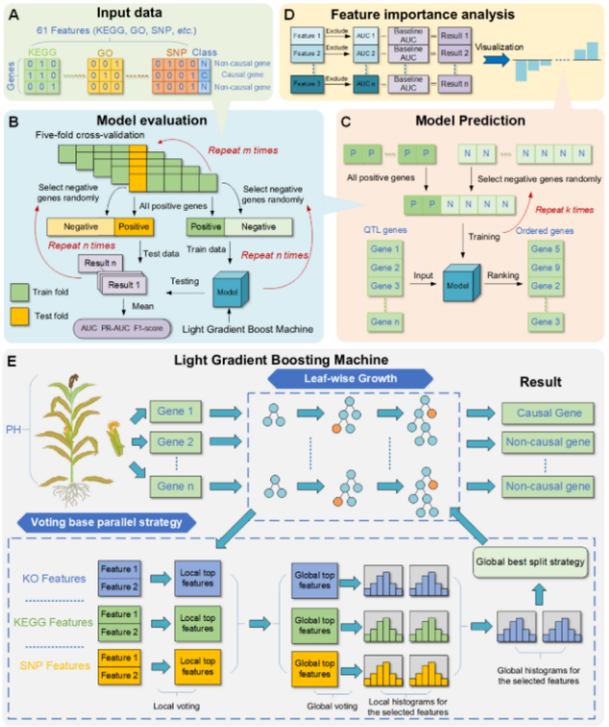

Arabidopsis Gene Knowledge Graph
AT1G13330 ——alias——> HOP2
├—annotated_with——> GO:...
├—associated_with_phenotype——> ...
├—expressed/observed_in——> floral organ / ...
├—interacts_with——> AT...
├—localized_in——> nucleus
└—supported_by——> PMID:...
GraphRAG
Tool Use
Trait
QTL
Candidate Gene List
LLM
Observed Phenotype Description
Result Report
"Causal Gene Ranking"
2
1
3

QTG_Finder (version 2.0)

QTG-Finder is a machine-learning pipeline to prioritize causal genes for QTLs identified by linkage mapping. We trained QTG-Finder models for Arabidopsis, rice, sorghum, Setaria viridis based on known causal genes and orthologs of known causal genes, respectively. By utilizing additional information like polymorphisms, function annotation, co-function network, paralog copy number, the models can prioritize causal genes for QTLs identified by QTL mapping.

Authors: Fan Lin, February 2020
Environment: Python=3.7.12 pandas=0.25.3 numpy=1.19.5 scikit-learn=0.21.2

QTG-LGBM: a method of prioritizing causal genes in quantitative trait loci in maize



vs

Benchmarking

Pheno2GeneRank

"Another Idea"

Trait

QTL

Candidate Gene List

Arabidopsis Gene Knowledge Graph



AT1G13330 ——alias——> HOP2
├—annotated_with——> GO:...
├—associated_with_phenotype——> ...
├—expressed/observed_in——> floral organ / ...
├—interacts_with——> AT...
├—localized_in——> nucleus
└—supported_by——> PMID:...

GraphRAG

Tool Use

QTG-LGBM: a method of prioritizing causal genes in quantitative trait loci in maize

LLM

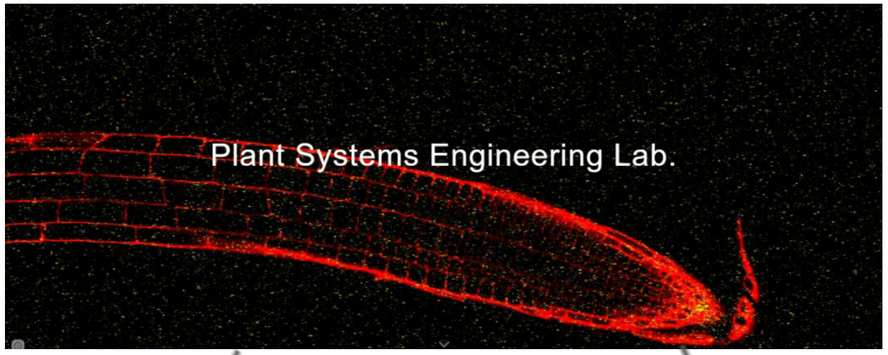
Observed Phenotype Description

Result Report

"Refinement of Causal Gene Ranking"



"Cowork"



Supervision
for Pheno2GeneRank

Plant System Research with Pheno2GeneRank



Systems Biology and Bioinformatics Laboratory

Pheno2GeneRank Development

1. Algorithm Design
2. Ablation Study

Evaluation

1. Metric Design
2. Test Dataset Construction
3. Validation